

$$4.2 \quad p(x) = a_1 x^3 + a_2 x^2 + a_3 x + a_0, \quad a_i \in \mathbb{R}$$

plati

$$p(0) = 1, \quad p(1) = 2, \quad p(2) = -3$$

ili $a_0^2 + \dots + a_3^2$ minimalni?

$$p(0) = 1: \quad a_0 = 1$$

$$p(1) = 2: \quad a_1 + a_2 + a_3 + a_0 = 2$$

$$p(2) = -3: \quad a_1 16 + a_2 8 + a_3 2 + a_0 = -3$$

izmatu SLR

$$\left(\begin{array}{cccc|c} 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 2 \\ 16 & 8 & 2 & 1 & -3 \end{array} \right) = (A|b)$$

a gledamo uktor koeficijenti $\vec{a}$ s minimalni normom, uči je shvatimo s gledati u najmanjše zbroje normog.

$$\text{Musi plati} \quad \vec{a} \in \text{Im } A^T, \quad A \vec{a} = b$$

$\Downarrow$

$$\vec{a} = A^T x$$

Dostizamo tedy

$$A A^T x = b \quad (\Rightarrow) \quad (A A^T | b)$$

$$A A^T = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 16 & 8 & 2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 16 \\ 0 & 1 & 8 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 5 & 5 \\ 1 & 31 & 31 \end{pmatrix}$$

$$(A A^T | b) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 5 & 5 & 2 \\ 1 & 31 & 31 & -3 \end{array} \right) \Rightarrow x = \frac{1}{40} \begin{pmatrix} 1 \\ 10 \\ -1 \end{pmatrix}$$

$$A^T x = \vec{a}$$

$$A^T x = \begin{pmatrix} 0 & 1 & 16 \\ 0 & 1 & 8 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix} \frac{1}{40} \begin{pmatrix} 1 \\ 10 \\ -1 \end{pmatrix} = \frac{1}{40} \begin{pmatrix} 1 \\ 5 \\ 5 \\ 1 \end{pmatrix}$$

$$\text{Polo} \quad (a_1, a_2, a_3, a_0) = \left(-\frac{3}{40}, \frac{1}{8}, \frac{3}{8}, \frac{1}{40}, 1 \right)$$

$$p(x) = -\frac{3}{40} x^3 + \frac{1}{8} x^2 + \frac{3}{8} x + \frac{1}{40} x + 1$$